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Journal of Magnesium and Alloys xxx (xxxx) xxx

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Thermodynamic and experimental evaluation of the sustainable recycling of magnesium alloy scrap by vacuum distillation based on vapor-liquid equilibrium

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Received 27 June 2023; received in revised form 15 October 2023; accepted 25 December 2023
Available online xxx

Abstract

Magnesium (Mg) alloys are widely used lightweight structural materials for automobiles and help reduce carbon emissions. However, their use increases the production of Mg alloy scrap, which is recycled at a much lower rate than aluminum, and its greater complexity poses challenges to existing recycling processes. Although vacuum distillation can be used to recycle Mg alloy scrap, this requires optimizing and maximizing metal recirculation, but there has been no thermodynamic analysis of this process. In this study, the feasibility and controllability of separating inclusions and 23 metal impurities were evaluated, and their distribution and removal limits were quantified. Thermodynamic analyses and experimental results showed that inclusions and impurity metals of separation coefficient $\lg\beta_i \leq -5$, including Cu, Fe, Co, and Ni below 0.001 ppm, could be removed from the matrix. All Zn entered the recycled Mg, while impurities with $-1 < \lg\beta_i < -5$ such as Li, Ca, and Mn severely affected the purity of the recycled Mg during the later stage of distillation. Therefore, an optimization strategy for vacuum distillation recycling: lower temperatures and higher system pressures for Zn separation in the early stage, and the early termination of the recovery process in the later stage or a continuous supply of raw melt can also prevent contamination during recycling. The alloying elements Al and Zn in Mg alloy scrap can be further recovered and purified by vacuum distillation when economically feasible, to maximize the recycling of metal resources.

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Keywords: Magnesium alloy; Scrap recycling; Thermodynamic analysis; Impurity removal; Vacuum distillation.

1. Introduction

Due to the ever-increasing global demand for reducing carbon emissions and sustainable development, magnesium alloys have attracted increasing attention as lightweight structural materials to reduce the energy consumption of vehicles. They are desired for their high specific strength and stiffness, castability, and low density and are 70% lighter than steel and 30% lighter than aluminum [1]. Mg alloys also have excellent biocompatibility, high hydrogen storage capacities, and other outstanding properties that give them unique advantages when used as functional materials in medical and new-energy applications [2,3].

The expansion of Mg alloy applications and increasing demands have produced large amounts of Mg alloy scrap, which must be recycled and reused. Furthermore, a variety of

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https://doi.org/10.1016/j.jma.2023.12.011
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advanced Mg alloys, Mg metal matrix composites (MMCs), and modified materials have recently been developed for different application environments to meet higher performance demands, as shown in Fig. 1. However, these new materials may contain potentially harmful elements as Co, Fe, and Ni, which complicate Mg recycling processes [4–7].

Across the Mg industry chain, primary Mg production has long been an energy-intensive process that generates greenhouse gases in amounts more than five times higher than the steel industry, as well as other pollution [8]. However, in contrast to other lightweight materials, magnesium and its alloys are theoretically fully recyclable, and recycling into secondary Mg requires only about 5% (less than 10.8 MJ/kg-Mg) of the energy required to produce it [9]. This approach can reduce resource consumption and carbon emissions while also providing attractive economic benefits [10]. Unfortunately, only a small proportion of Mg alloy scrap is recycled, while the rest is incinerated or buried. The International Magnesium Industry Association conducted a comprehensive survey of the material flow of Mg and found that the end-of-life recy-

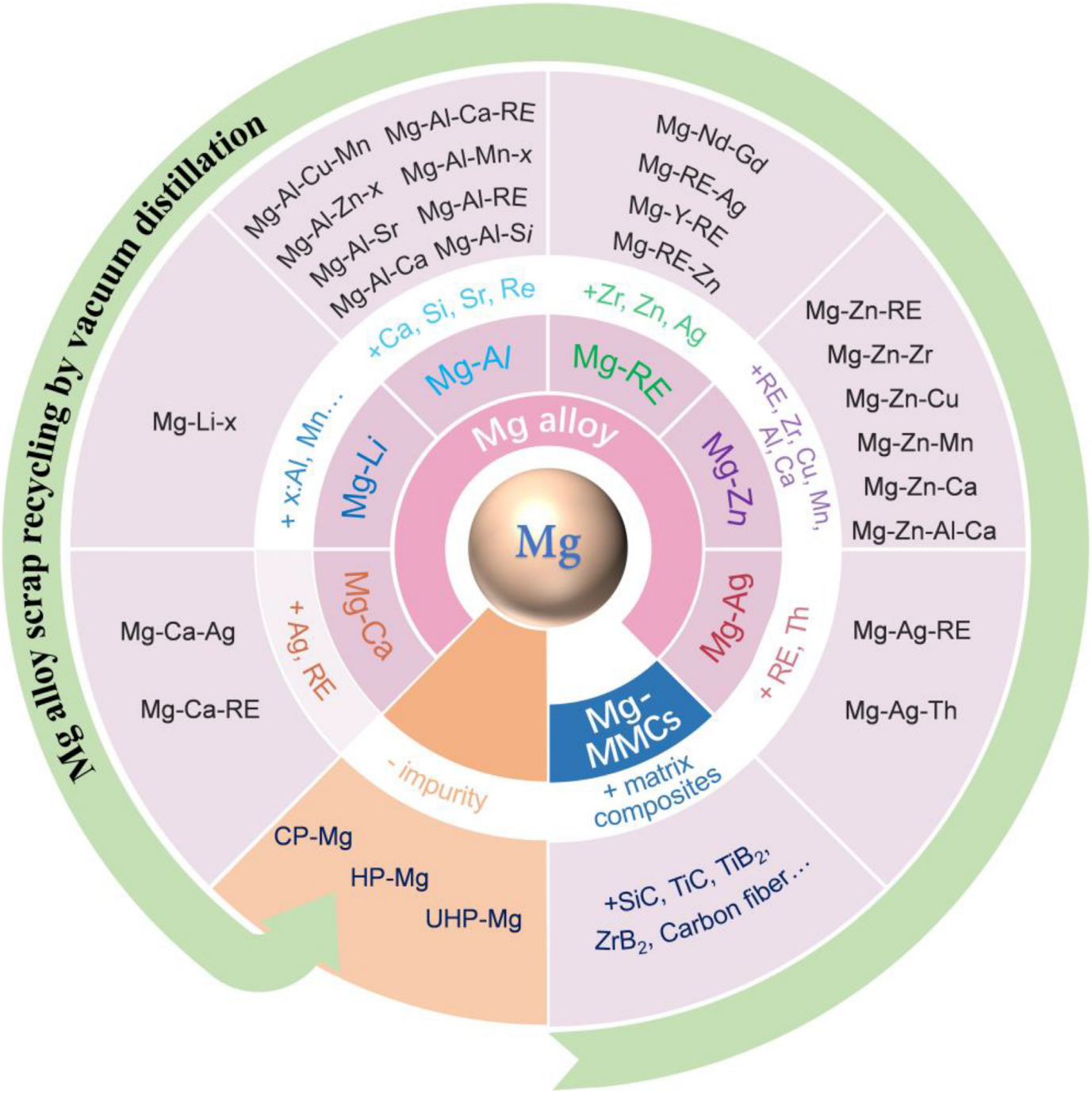


Fig. 1. Summary of Mg alloy development [4].

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cling input rate for magnesium in the European Union was 7%, which is below the global average of 12% for aluminum [11,12]. Mg may have a low recycling rate because the dispersion of Mg alloy products leads to a low collection rate, or because the recovery of mixed complex Mg alloy scrap from shredders is inefficient and costly [13]. Additionally, anti-corrosion coatings on end-of-life Mg alloy products contain harmful elements that pose greater challenges to current recycling processes [13,14]. Therefore, a sustainable method to recycling complex scrap is necessary to improve the recovery of Mg.

The most important step in the recycling of Mg alloy scrap is the elimination of inclusions and impurities [15]. Two main recycling methods are based on flux and fluxless refining [16]. Flux refining recycling is a relatively low-cost traditional technology for removing inclusions, and fluxless recycling further removes smaller inclusions and impurities in the melt to obtain higher-quality recycled Mg alloys. However, both methods face challenges when used to process complex scrap, such as their insufficient applicability to recovered materials, contamination by corrosive vapors and protective gases, and difficulty in removing hazardous impurity elements. Vacuum distillation recycling is widely applicable to many raw materials and has a high efficiency, sustainability, generates no pollution, and is the most effective method for removing harmful impurity elements such as Fe, Cu, and Ni. The recycled products are pure Mg or Mg alloys [17].

Thermodynamic analysis has been widely used to evaluate metal waste recovery processes for steel, copper, lead, zinc, aluminum, titanium, cobalt, and nickel [18–22]. It provides guidance on process route selection and useful information for the economical recycling of secondary metal resources. Before recycling Mg alloy scrap by vacuum distillation, especially complex scrap from shredder plants, it is necessary to thermodynamically analyze the recycling process. This helps clarify the removability and controllability of the impurity components in the Mg alloy scrap, but few reports have done this. Gesing and Das [23] determined the electrolyte composition, a crucial key parameter of the RE12^{TM} process for the electrorefining Mg from Al scrap by thermodynamic analysis. This greatly reduced the process development time and provided reliable material composition and reactivity predictions for the products. Jung et al. [24] studied the complex chemical reactions occurring during the flux recycling of Mg alloy scrap and showed that the main non-metallic inclusion MgO was removed via physical adsorption instead of a chemical reaction. They also showed the limitations the lost Ca, Sr, and rare-earth elements by burned and could not remove Cu, Ni, and Sn. Takehito et al. [5] further evaluated the possibility of using oxidation and flux chlorination to remove 25 impurity metals during Mg recycling. The conclusions indicated that reactive elements such as alkali metals, alkaline earth metals, and rare earths could be removed by oxidization or chlorination with salt flux. Other elements were hardly removed, which highlights the difficulty and challenges of removing impurity metals from Mg alloys during recycling. It also showed that distillation-based recovery methods of Mg from Mg alloy are theoretically feasible, but there is a lack of fundamental thermodynamic support.

In this study, based on the foresight of complex mixed magnesium alloy scrap, we evaluated the possibility and separation capacity of removing inclusions and 23 impurities during vacuum distillation recycling process by considering all relevant thermodynamic parameters. A removal limit of metal impurities during the recycling process and a quantitative analysis of the distribution of impurity metals in the vapor-liquid phase on purity were performed, which provide a further understanding of the behavior of impurities separation and removal during vacuum distillation recycling. This fills in the gaps of thermodynamic criteria and key parameters on vacuum distillation recycling complex Mg alloy scrap, provides a crucial guide to optimize the recycling process and obtain a consistent product quality during actual industrial processes to improve the sustainable circulation of Mg metal in the industry chain.

2. Thermodynamic method

2.1. Saturation vapor pressure

Vacuum distillation technology takes advantage of the difference in saturation vapor pressure of different pure substances for alloy separation or metal purification. Substances with higher saturation vapor pressures are preferentially volatilized into the vapor phase and then condensed, while others remain in the residue. Hence, the difference between the saturation vapor pressure of different substances can be used to roughly estimate whether they can be separated. The saturation vapor pressure of substances can be calculated by formula (1), which is a function of temperature [25]:

$$\lg P_i^* = \mathrm{A}T^{-1} + \mathrm{B}\lg T + \mathrm{C}T + \mathrm{D} \quad (1)$$

where P_i^* is the saturation vapor pressure of the pure substance i; The parameters A, B, C and D are the evaporation constants for substance i; T is the absolute temperature in K.

2.2. Stability of inclusions

Non-metallic inclusions may decompose under high temperature and vacuum conditions, which will affect the purity of recycled Mg. Inclusions decomposition trends can be judged by the Gibbs free energy under real conditions ($\Delta_r G$) using formula (2). A higher $\Delta_r G$ indicates that the decomposition reaction is less likely to proceed and vice versa.

$$\Delta_r G = \Delta_r G^o + \mathrm{R}T\, lnQ \quad (2)$$

where R is the gas constant; T is the absolute temperature in K; $\Delta_r G°$ is the standard Gibbs free energy change of a reaction; Q is the ratio of pressure or concentration of the substance before and after the reaction under actual conditions.

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2.3. Separation coefficient β

During material preparation and metallurgy, actual Mg melts cannot be regarded as an ideal solution due to complex interactions between elements. This is the reason for the positive and negative deviations between the actual vapor pressure and the saturation vapor pressure of a pure substance.

Therefore, to accurately represent the thermodynamic behavior of the solution system in thermodynamic calculations, the activity must be used instead of concentration. To more accurately evaluate the separation capacity of between the components in an alloy, the separation coefficient β is introduced based on the activity of impurities [25]. For Mg-*i* binary alloys, the separation coefficient criterion is expressed as Formula (3).

$$\beta_i = P_i^* \gamma_i \ / \ P_{Mg}^* \gamma_{Mg} \tag{3}$$

where P^*_i and P^*_{Mg} are the saturation vapor pressures of pure substances i and Mg, respectively; γ_i and γ_{Mg} are the activity coefficients of impurity i and Mg at a given temperature, respectively; β_i is the separation coefficient of impurity i, where a value of β_i closer to 1 indicates that it is more difficult to separate an impurity from the Mg solvent. Otherwise, impurity i will evaporate into the vapor phase ($\beta_i > 1$) before Mg or remain in the melt ($\beta_i < 1$) until Mg is completely evaporated.

The activity coefficients of impurity elements and Mg in a Mg-*i* binary system at any temperature and composition can calculated by Formulas (4) and (5), derived from the Redlich–Kister polynomial [26]. This study concentrated on the recycling of the main metal Mg without considering interactions between alloying elements, and the corresponding discussion has been limited to Mg–*i* binary alloys.

$$RT \ln \gamma_i = x_{Mg}^2 \left\{ {}^0L_{Mg-i} + \sum_{k=1}^{n} {}^kL_{Mg-i}(x_{Mg} - x_i)^{k-1} \times \left[(2k+1)x_{Mg} - (2k+1)\right] \right\} \tag{4}$$

$$RT \ln \gamma_{Mg} = x_i^2 \left\{ {}^0L_{Mg-i} + \sum_{k=1}^{n} {}^kL_{Mg-i}(1-2x_i)^{k-1} \times [(2k+1) - 2(k+1)x_i] \right\} \tag{5}$$

where x_{Mg} and x_i are the mole fractions of Mg and impurity i, the initial mole fraction of impurity metals in the Mg solvent was fixed at 1 % (x_i=0.01). ${}^kL_{Mg\text{-}i}$ is the k-th binary interaction parameter between Mg and element i, k denotes the order and is generally in the range of 0–3. These L parameters may depend on temperature as $L = A + BT + CT \ln T$, where the constants A, B, and C were obtained from previous studies. T is the recycling temperature, which was set to 973 K based on the melting point of Mg, viscosity, other physical properties, and actual industrial settings.

2.4. The vapor-liquid phase equilibrium (VLE) diagram

To quantitatively estimate the removal limit and content of each component in the vapor and liquid phases, knowing the equilibrium composition is critical [27]. Therefore, the VLE composition diagram (x-y diagrams) was introduced to accurately quantify the distribution of alloy components between vapor-liquid phases during vacuum distillation.

When the impurity i in the Mg melt reaches thermodynamic equilibrium between the vapor and liquid phases, the relationship of impurity concentrations between in the vapor phase (g) and the liquid phase (l) is given by the Formula (6) [28]:

$$\omega_{i,g} = \left(1 + \frac{x_{Mg,l}}{x_{i,l}} \cdot \frac{M_{Mg}}{M_i} \cdot \frac{\gamma_{Mg}}{\gamma_i} \cdot \frac{P_{Mg}^*}{P_i^*}\right)^{-1} \tag{6}$$

where $\omega_{i,g}$ is the mass fractions of impurity i in the vapor phase; $x_{Mg,l}$ and $x_{i,l}$ are the mole fraction of Mg and i in the liquid phase; M_{Mg} and M_i are the molar mass of Mg and i, respectively. For major elements such as Al, Zn, Ca, and Li, the initial concentration of impurity i in Mg was assumed to be in the range of 1–99 wt.%. For minor elements such as Cu, Fe, Co, and Ni, their concentration in Mg was assumed to be in the range of 1–10 000 ppm (i.e., 1 ppm –1 wt.%).

To simply and rapidly determine the quantitative relationship between product composition and temperature and pressure during vacuum distillation, the vapor-liquid equilibrium phase diagram was introduced. The expression for a binary alloy system of Mg-*i* is shown in Eqs. (7) and (8) [28]:

$$x_i = \ (P - P_{Mg}^* \gamma_{Mg}) \ / \ (P_i^* \gamma_i - P_{Mg}^* \gamma_{Mg}) \tag{7}$$

$$y_i = P_i^* \gamma_i x_i / P \tag{8}$$

$$\text{Liquid phase} : x_i + x_{Mg} = 1 \tag{9}$$

$$\text{Vapor phase} : y_i + y_{Mg} = 1 \tag{10}$$

where the x_i and y_i are the mole fraction of impurity i in the liquid phase and vapor phase, determined from equations (9) and (10); P is the total pressure of the Mg−i binary system; P^*_{Mg} and P^*_i are the saturation vapor pressures of Mg and impurity i, respectively; γ_{Mg} and γ_i are activity coefficients of Mg and impurity i in an Mg−i binary alloy, respectively.

The *P-x(y)* (pressure-composition) and *T-x(y)* (temperature-composition) phase diagrams of the Mg-*i* system were obtained by substituting the corresponding activities of different components at different temperatures into Eqs. (7)-(10) and performing a series of iterative calculations [29].

3. Results and discussion

Inclusions and metal impurities are inevitably introduced during the actual production and recovery of Mg alloys. These inclusions will degrade the mechanical properties and corrosion resistance of Mg alloy structural products [30], making

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it critical to control inclusions to promote the widespread use of Mg and its alloys [31]. Even small amounts of impurity metals can significantly reduce the corrosion resistance of Mg alloys, especially Cu, Fe Co, and Ni [32].

3.1. Thermodynamic analysis of non-metallic inclusions

Inclusions in Mg alloys can be divided into non-metallic and intermetallic inclusions (iron-rich phases, e.g., $Al_8(Mn,Fe)_5$, α-AlMnFe, α-Fe, and $Fe_3(Al,Si)$). Lun Sin presented a detailed review of the types and characteristics of inclusions in Mg alloys [30]. This study focused on the thermodynamic behavior of non-metallic inclusions during vacuum distillation recycling, while iron-rich intermetallic inclusions were not discussed because they were mainly composed of Fe, Mn, Al, Si, and other less-volatile elements that remained in the residue. Non-metallic inclusions commonly exist in the melt, including oxides (mainly MgO, FeO, Fe_2O_3, Al_2O_3), nitrides (Mg_3N_2), chlorides ($MgCl_2$, KCl, NaCl, $CaCl_2$), sulfides (MgS), and fluorides (MgF_2). These are primarily caused by reactions with air or a protective atmosphere and by flux residue [33]. Most inclusions are oxides and chlorides [30,34], and one study showed that MgO accounts for over 80% of all Mg alloy inclusions [35]. In addition, coatings to extend a product's service life can introduce other impurity metals, as well as non-alloying metal oxides, and all alloying elements have the possibility of becoming oxides in the anodizing and ceramic coating process [14].

Thermal stability analysis of non-metallic inclusions showed that the Gibbs free energy of their decomposition was positive and large even in the conditions of high temperatures and vacuum (10 Pa), as shown in Fig. 2(a). This indicates that these common non-metallic inclusions should not decompose during the vacuum distillation and remain in the residue without affecting the quality of recycled Mg.

Chlorides are volatile but significantly less volatile than Mg, as shown in Fig. 2(b). Chloride inclusions cannot affect recycled Mg because their content in Mg alloys is strictly controlled at less than 500 ppm [30]. For die-cast Mg alloys, the content of a thin film and particle inclusions inside an alloy must be below 100 cm^2/kg and 100 mm^3/kg. For thin-walled castings, the restrictions are even more stringent [36,37]. Moreover, all alloying oxides can be reduced by Mg to their corresponding metal and MgO. Therefore, this situation will be discussed later during the thermodynamic analysis of impurity metals.

3.2. Thermodynamic analysis of impurity metals

Adding various alloying elements improves the corrosion resistance of Mg, thus preparing Mg alloys with different properties that meet the requirements of their respective applications as structural or functional materials [38]. In contrast to alloying, the use of vacuum distillation to recycle Mg can be viewed as a separation and purification process, in which almost all of the metallic elements need to be separated from Mg. During recycling, relative to the metal Mg solvent, other

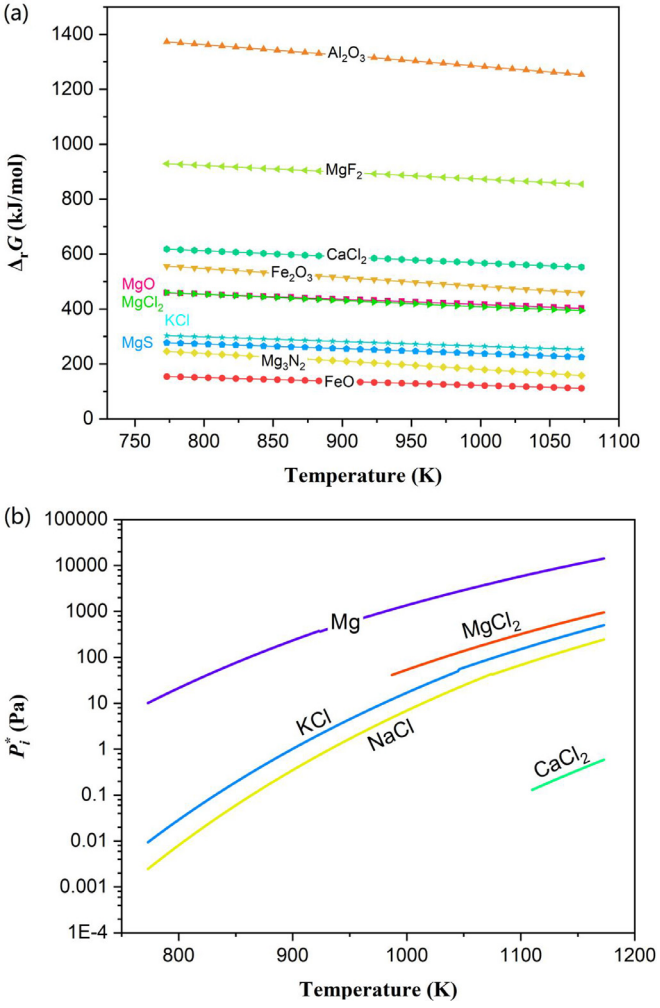


Fig. 2. (a) Gibbs free energy (assuming P_{system}=10 Pa, $\Delta_r G=\Delta_r G°+RT\ln Q$) of decomposition of possible non-metallic inclusions in Mg alloys; (b) Saturation vapor pressure of some chloride inclusions.

metals including alloying elements are considered to be impurity metals, which facilitates the classification and analysis in this study.

3.2.1. Saturation vapor pressures and *separation coefficient* β

The saturation vapor pressure curves versus temperature for different elements in Mg alloy scrap were obtained at 773–1073 K using Formula (1), as shown in Fig. 3(a), but the data of Zr, Cr, Ce, Gd, and Nd in this temperature range were incomplete. Hence, Fig. 3(b) further was further drawn to shows the saturation vapor pressures of 25 different elements including the above elements in Mg alloy scrap at 973 K. It follows that the evaporation order of metals was K > Na > Zn > Mg > Li > Sb > Sr > Ca > Bi > Pb > Mn > Ag > Al > Sn > Cu > Nd > Cr > Fe > Gd > Co > Ni > Y > La > Ce > Si > Zr.

Under an operating temperature of 973 K, impurity metals were divided into three categories according to their different saturation vapor pressures in Fig. 3(b). The role of these elements in Mg alloys and their range of additions were fur-

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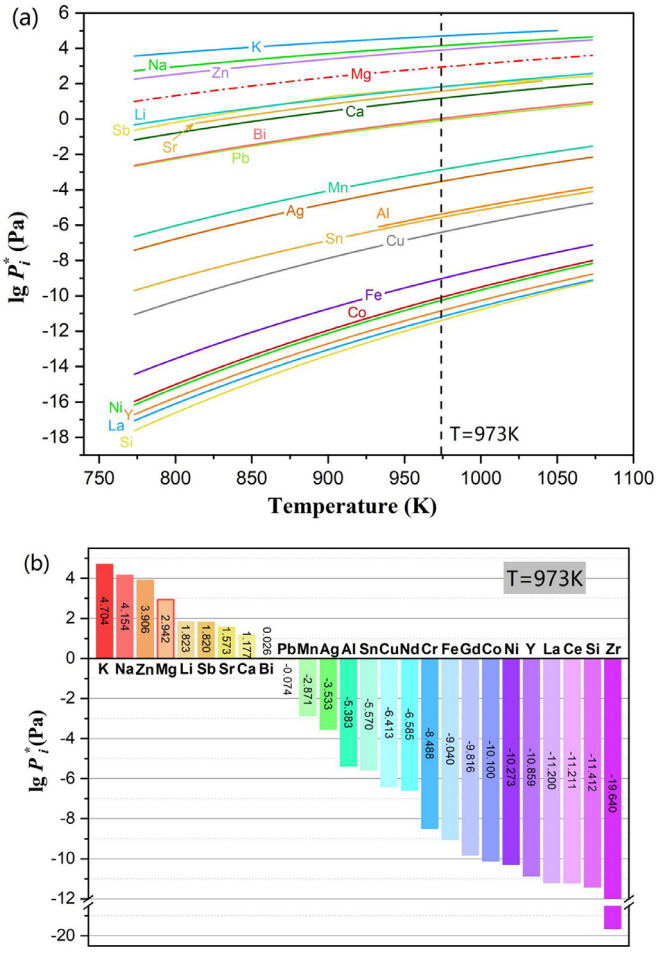


Fig. 3. Saturation vapor pressure of elements in the Mg alloys scrap at temperature: (a)773–1073 K; (b)973 K.

ther explained to provide basic data to support subsequent thermodynamic analysis.

(i) Saturation vapor pressures higher than that of Mg: The saturation vapor pressure of K, Na, and Zn were at least an order of magnitude greater than that of Mg, implying that these metals volatilized into the vapor phase before Mg and thus affected the purity of recycled Mg. Generally, the contents of K and Na in Mg alloy are negligible and were only taken into consideration when manufacturing high-purity or ultra-high purity Mg (purity > 5 N) [39]. An example of adding 0.1 wt.% Na to age-hardened Mg-Sn-Na alloys has been reported [40].

Among them, focus should be placed on Zn, as it is another efficient alloying element except aluminum and is widely utilized in Mg-Zn alloys [41], in which the Zn content in ZE90M can reach 9 wt.% [42]. However, the high volatility and wide application of Zn pose great challenges to the vacuum distillation recycling of complex mixed Mg alloy scrap. Because the Zn content can fluctuate drastically with the scrap composition, its content in recycled Mg is strictly limited, which causes problems for the reuse and subsequent disposal of recycled Mg.

(ii) Saturation vapor pressures slightly lower than that of Mg (including Li, Sb, Sr, Ca, Bi, and Pb): The saturation vapor pressures of Li, Sb, Sr, and Ca were 1–2 orders of magnitude lower than that of Mg, while that of Bi and Pb were about 3 orders of magnitude lower. However, because the saturated vapor of these elements was not sufficiently different from that of Mg, the recovery of Mg may still be affected during the distillation process. Li and Ca metals are especially important alloying elements for Mg alloys and have been used to develop creep-resistant and degradable Mg-Ca alloys, as well as superlight Mg-Li alloys for aviation applications [43,44]. Trace amounts of Sb, Sr, Bi, and Pb are often added to Mg alloys as alloying elements, among which Bi, Sb, and Sr enhance the creep resistance of alloys [45,46], and ~1 wt.% Pb can inhibit the corrosion of Mg alloys [47].

(iii) Saturation vapor pressures much lower than that of Mg: Among the 23 impurity metals except for K and Na that may exist in Mg alloy scrap, 16 impurity metals displayed saturation vapor pressures much lower than those of Mg by more than 5 orders of magnitude. This means that these elements remained in the liquid phase and showed great separation potential. Among them, Mn, Ag, Sn, Al, Zr, and rare earth elements (Nd, Gd, La, Ce, Y) are all important alloying elements in different series of Mg alloys. In particular, Al is the most widely used alloying element in Mg alloys, and 9 wt.% is added to the most widely used AZ91 alloy, and a maximum of 10 wt.% is added to AM100A [48]. Due to increasing market demands for high-performance Mg alloys, the consumption of rare earths (Nd, Gd, La, Ce, and Y) in Mg alloys has also increased. The amount of added rare earths varies with each alloy brand. For example, WV115Z contains 16 wt.% of rare earth metals [48].

The most harmful impurity metals are Cu, Fe, Co, and Ni, even small amounts of which can make Mg structurally unstable [17]. Therefore, their contents in Mg alloys are strictly limited to 35–50 ppm for Fe, 20–50 ppm for Ni, and 100–300 ppm for Cu (wt.%) [49]. Although Co is uncommon in Mg alloys and there is no requirement for its content, but it must be taken into account during the recovery of complex scrap since its harmfulnesss as same as Cu. The saturation vapor pressure of these impurity metals is more than 10 orders of magnitude lower than that of Mg, which will facilitate their removal. Furthermore, the recycled Mg must maintain strict impurity limits to enter the Mg metal cycle smoothly.

The logarithm of impurity metal separation coefficients at 973 K is given in Fig. 4 to clearly demonstrate the separation capacity of each impurity from Mg. The relevant thermodynamic parameters for this study are also listed in Table 1, where $\beta_{Zn} = 2.398$ ($\lg\beta_{Zn} = 0.38$), which is close to unity, indicating the difficulty of separating Zn and Mg. These results agree with the experimental observation that Zn is the most difficult impurity to remove from Mg [71]. A comparison of the results in Fig. 3 and Fig. 4 shows a discrepancy between the separation coefficient and the saturation vapor pressures ratio, which was caused by variations in the activities of impurity metals in Mg melt. The activity values of Bi and Sb in Mg melt were very small due to strong interactions

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Table 1
Separation coefficient of impurity metals (x_i = 0.01) from solute magnesium at 973 K and related thermodynamic parameters used in this study.

Element	ASTM Codes	P^*(973 K)	Ref.	γ_i	γ_{Mg}	Ref.	β_i
Mg	–	8.755×10^{2}	[50]	–	–	–	–
Ag	Q	2.928×10^{-4}	[50]	0.0007	0.9994	[52]	2.356×10^{-10}
Al	A	4.143×10^{-6}	[50]	1.4035	1.0003	[53]	6.640×10^{-9}
Bi	B	1.063	[50]	0.0003	0.9989	[54]	3.366×10^{-7}
Ca	G	1.503×10^{1}	[50]	0.0379	0.9993	[55]	6.514×10^{-4}
Ce	E	6.157×10^{-12}	[51]	0.0021	0.9983	[56]	1.512×10^{-17}
La	E	6.311×10^{-12}	[50]	0.0031	0.9985	[57]	2.239×10^{-17}
Nd	E	2.597×10^{-7}	[51]	0.0029	0.9987	[58]	8.556×10^{-13}
Cr	R	3.250×10^{-9}	[50]	1.0000	1.0000	–	3.712×10^{-12}
Co	–	7.938×10^{-11}	[50]	8.2723	1.0002	[59]	7.499×10^{-13}
Cu	C	3.865×10^{-7}	[50]	0.0514	0.9999	[53]	2.271×10^{-11}
Fe	F	9.130×10^{-10}	[50]	2759.993	1.0009	[53]	2.876×10^{-9}
Gd	V	1.529×10^{-10}	[51]	0.0014	0.9985	[60]	2.500×10^{-16}
Li	L	6.658×10^{1}	[50]	0.1488	0.9999	[61]	1.132×10^{-2}
Mn	M	1.347×10^{-3}	[50]	44.7979	1.0003	[62]	6.889×10^{-5}
Ni	N	5.334×10^{-11}	[50]	0.0157	0.9995	[63]	9.552×10^{-16}
Pb	P	8.426×10^{-1}	[50]	0.0118	0.9993	[64]	1.135×10^{-5}
Sb	Y	6.601×10^{1}	[50]	5.10×10^{-6}	1.0020	[65]	3.838×10^{-7}
Si	S	3.875×10^{-12}	[50]	0.00318	1.0002	[66]	1.409×10^{-17}
Sn	T	2.691×10^{-6}	[50]	0.0261	0.9996	[67]	8.034×10^{-11}
Sr	J	3.744×10^{1}	[50]	0.0635	0.9995	[68]	2.72×10^{-3}
Y	W	1.384×10^{-11}	[50]	0.0146	0.9993	[69]	2.312×10^{-16}
Zn	Z	8.058×10^{3}	[50]	0.2605	0.9999	[70]	2.398
Zr	K	2.290×10^{-20}	[28]	1.0000	1.0000	–	2.616×10^{-23}

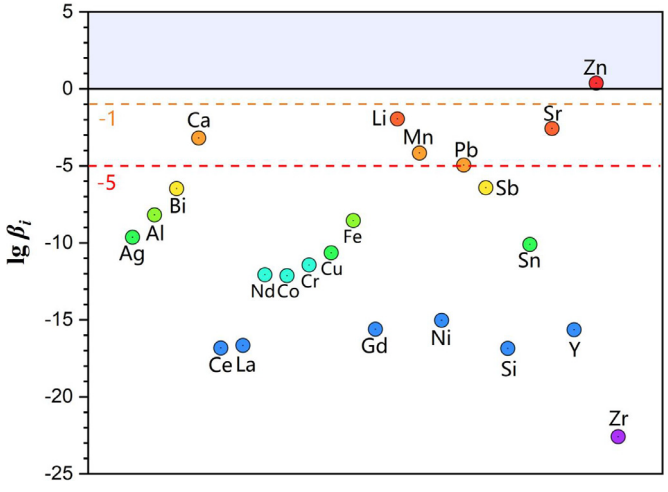


Fig. 4. Separation coefficients of impurity metals at 973 K.

between them and magnesium to form stable compounds of Bi_2Mg_3 and Sb_2Mg_3 from their phase diagram, resulting in a larger separation coefficient, indicating they were easier to separate. In contrast, Fe and Mn showed high activity values due to their extremely low solubility in Mg and did not form any compounds. The results unexpectedly showed that Mn was more difficult to separate from Mg than Pb. There was no activity data for Cr and Zr, so their activity values were set to 1 due to their low solubility in Mg.

The separation coefficient is a crucial thermodynamics criterion for evaluating the separation and removal capacity of metal impurities from Mg melt, and this criterion is $\lg\beta_i = -5$ for Mg alloys. When $\lg\beta_i \leq -5$, most impurities can be removed from scrap (including Bi, Sb, Al, Fe, Ag, Sn, Cu, Cr, Co, rare earths, Ni, Si, and Zr). When $\lg\beta_i > -5$, it is difficult to separate impurities such as Ca, Li, Mn, Pb, and Sr. Furthermore, $\lg\beta_i \geq -1$, which shows that it is almost impossible to remove impurity Zn via vacuum distillation.

3.2.2. VLE composition diagrams

To better analyze the limit and effect of impurities removal, major alloying elements (Zn, Li, Ca, and Al) and harmful elements (Cu, Fe, Co, and Ni) were selected as examples.

For major alloying elements, Zn exhibited distinct behavior compared with other elements (Fig. 5), where the vapor phase content of Zn exceeded the liquid phase content over all ranges. Further, the Zn content in the vapor phase was almost 2.5 times higher than that of the liquid phase, particularly at ~20 wt.% of the liquid phase content, as shown in Fig. 5(a), because Zn has a higher saturation vapor pressure than Mg, in the case of low zinc content in liquid phase, a small increase of Zn in the liquid phase will significantly increase the proportion (i.e., content) of Zn in the vapor phase. It can be inferred that Zn began to evaporate from the melt during the early recycling stage before Mg underwent significant volatilization. Another difference is that the Zn content in the liquid phase decreases as distillation proceeds, expressed in Fig. 5(a) is that the Zn content curve shifted from right to left due to the high volatility of Zn, and due to it evaporate into the vapor phase, and thus Zn in the liquid phase was consumed. It should also be noted that the Zn content in the vapor phase decreased as the temperature increased, indicating that a lower temperature was more favorable for Zn volatilization and separation from Mg. This indicates that during heating, zinc began to volatilize before the tempera-

ture reached the recovery temperature. Therefore, zinc was removed first at this stage. In addition, zinc could also be collected during this process to theoretically achieve zinc recovery and recycling.

Li, Ca, and Al were all enriched in the liquid phase, so they consistently showed lower contents in the vapor phase in all ranges. For example, even when their content in the liquid phase reached 20 wt.% at 973 K, the Li content in the vapor phase was only 0.34 wt.%, Ca was 0.10 wt.%, and Al was 0.0009 ppm. In addition, their contents in the vapor phase increased with the temperature, which is consistent with the saturation vapor pressure trend. Compared with Li and Ca, Li exhibited greater volatility than Ca over all ranges at the same temperature. When the vapor phase contents of Li and Ca both reached 10 ppm at 973 K, their contents in the liquid phase were 10 times different, with values of 0.1 wt.% and 1 wt.%, respectively. Due to the continuous evaporation of Mg during distillation, these alloying elements were enriched in the liquid phase, resulting in a rapid increase in their vapor phase content during the later stage of distillation. They even fully entered recycled Mg, as shown in Fig. 5(b) to (d), which further reduced the purity of recycled Mg. The critical points of the liquid phase content were obtained by using the wire-drawing method in the VLE diagrams. The impurity content in the vapor phase increased rapidly after its liquid phase content reached a critical value. For example, at 973 K, when the liquid phase contents of Li and Ca reached their critical values of 57.42 wt.% and 75.32 wt.%, their vapor phase contents were about 13 wt.%, after which they increased rapidly. The rate of change of the vapor phase content of Ca was twice that of Li.

For the primary alloying element Al, a higher temperature caused a rapid increase in its vapor phase content, especially at a high liquid phase content. For example, when its liquid phase content was 90 wt.%, its vapor contents were 0.076 ppm, 0.20 ppm, and 0.50 ppm at 973 K, 1023 K, and 1073 K. In addition, even when the Al content in the liquid phase reached 99 wt.%, the Al content in the vapor phase was only 1 ppm, indicating that Al could be separated well from Mg.

The above analysis showed that it is necessary to improve the volatilization of Zn and inhibit Mg volatilization during the initial low-temperature stage, so that Zn is completely volatilized before large amounts of Mg begin to evaporate.

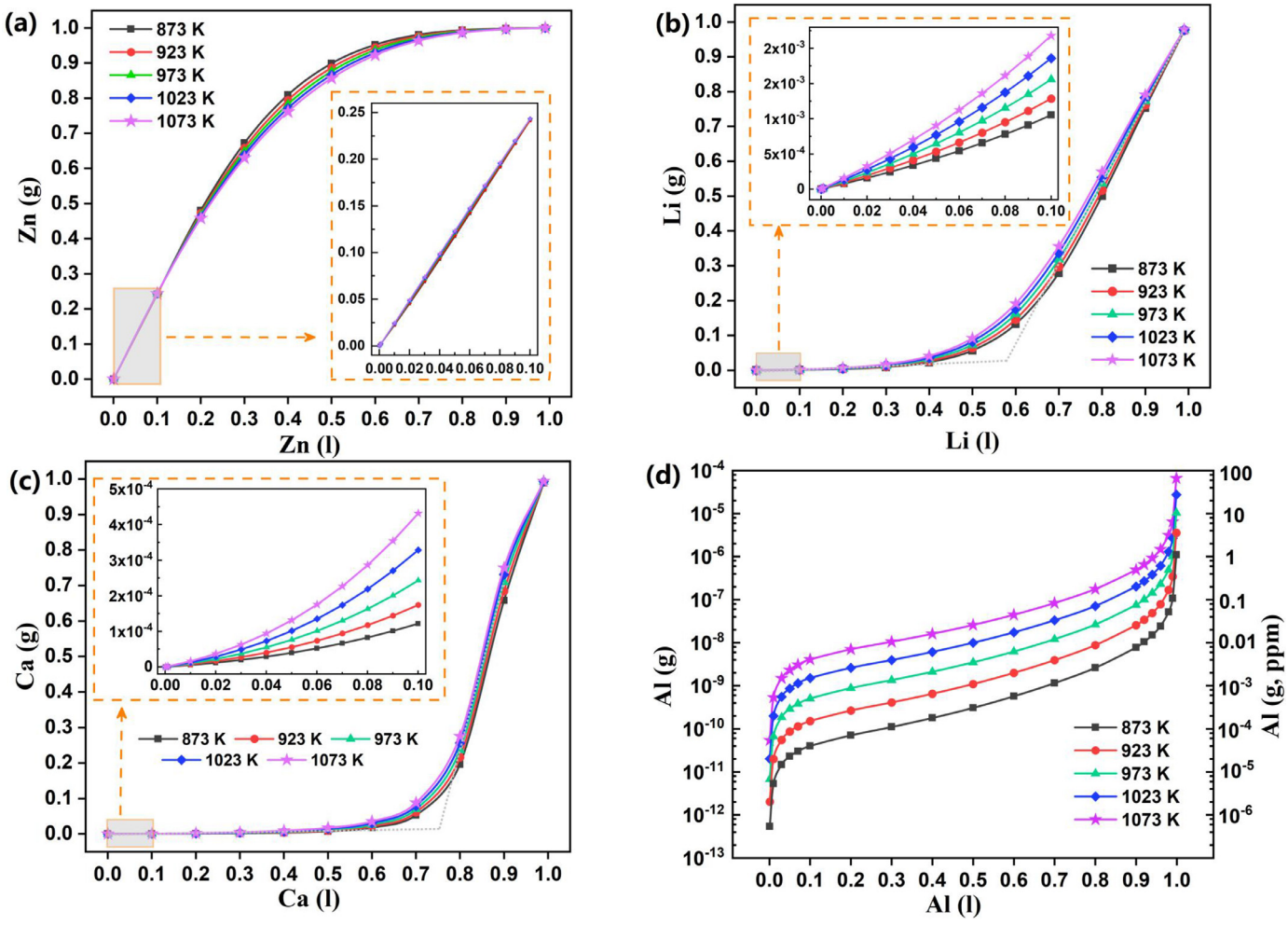


Fig. 5. VLE composition diagrams of impurities (a) Zn, (b) Li, (c) Ca, (d) Al, (e) Cu, (f) Fe, (g) Co, (h) Ni in Mg-*i* binary system.

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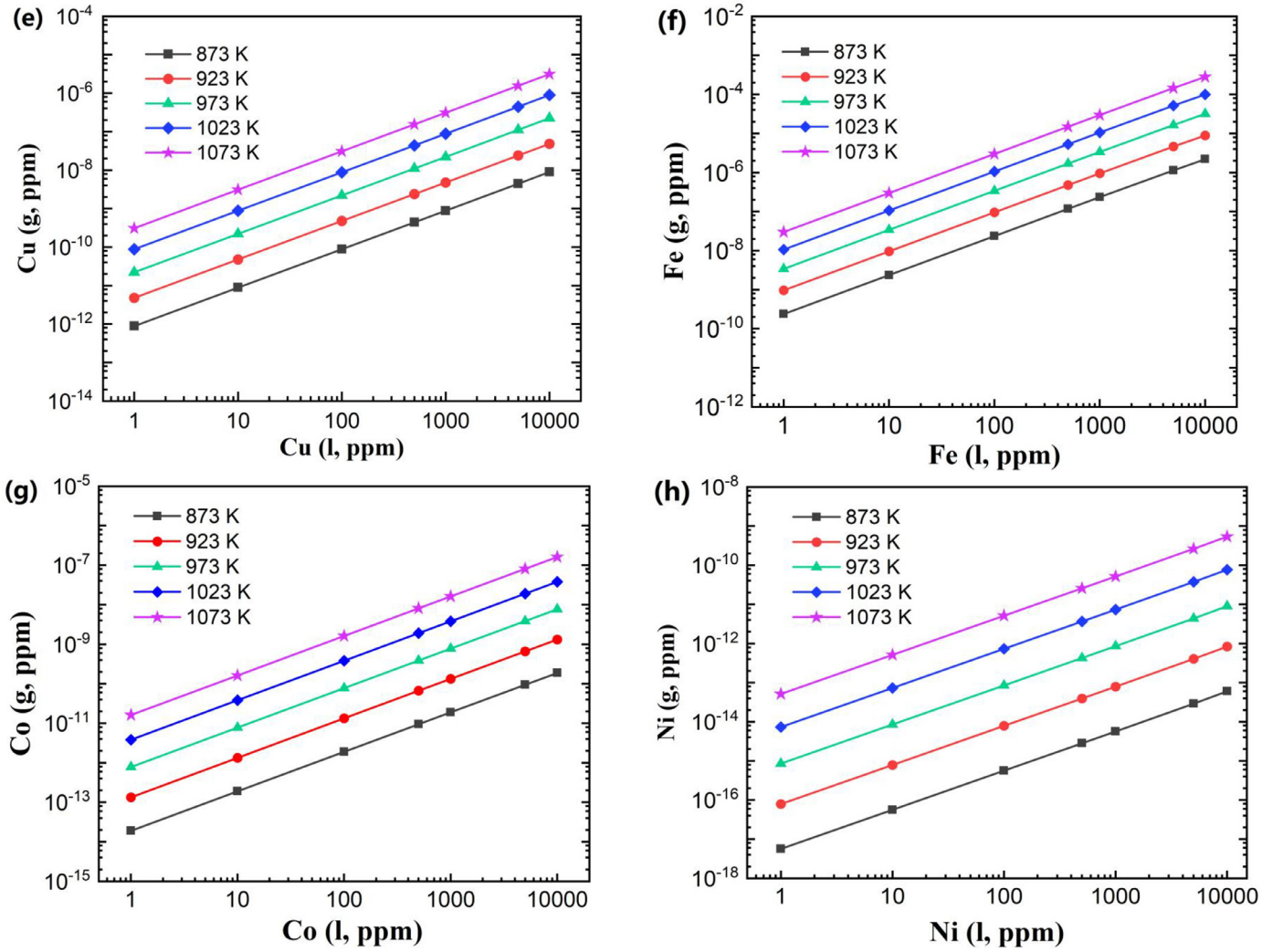


Fig. 5. Continued

During the late stage of distillation, to avoid contamination by impurity metals enriched in the liquid phase into the regenerated Mg, it should be finished before, or continuous feeding should be adopted to maintain a low content of impurities in the melt.

The problem with recovering old scrap from end-of-life products is that uncontrolled pollution during recycling will increase the contents of harmful elements. For this reason, the impurity content range of 1–10 000 ppm was considered in this study. The vapor phase contents of four harmful trace metals (Cu, Fe, Co, and Ni) increased with the temperature and their liquid phase content, as shown in Fig. 5(e)-(f). Under the same conditions, the vapor phase contents of Cu, Co, and Ni followed the same order as the saturation vapor pressure: Cu > Co > Ni. However, the vapor phase content of Fe was significantly higher than that of Cu, which was related to the higher activity of Fe due to its low solubility in Mg (the maximum solid solubility of Fe was 0.001 wt.% [4,71]). Analysis of the VLE composition diagrams showed that the vapor phase content was less than 0.001 ppm, even when the liquid phase content of the four impurities reached 1 wt.%, which was far below the limit of high-purity magnesium for these impurities [39]. Therefore, vacuum distillation can remove harmful impurities such as Cu, Fe, Co, and Ni, which are impossible to remove using other recycling processes, to obtain recycled Mg with low or even ultra-low contents of these harmful impurities.

3.2.3. VLE phase diagram

Vacuum distillation is highly dependent on controlling the system pressure and temperature. Moreover, the equilibrium distribution of each element in the vapor and liquid phases and its relationship with temperature and system pressure can be understood in real-time via the VLE phase diagram, which can provide significant information for process optimization and understanding impurities behavior. The analysis of the most crucial Mg-Al and Mg-Zn systems in Mg alloys is shown in Fig. 6. It is worth noting that the triple point of Mg is located at 923 K and 333 Pa [25], which implies that the discussion of the VLE phase diagram is only meaningful if the system pressure is greater than 333 Pa to ensure the formation of a liquid phase. Otherwise, Mg will sublimate.

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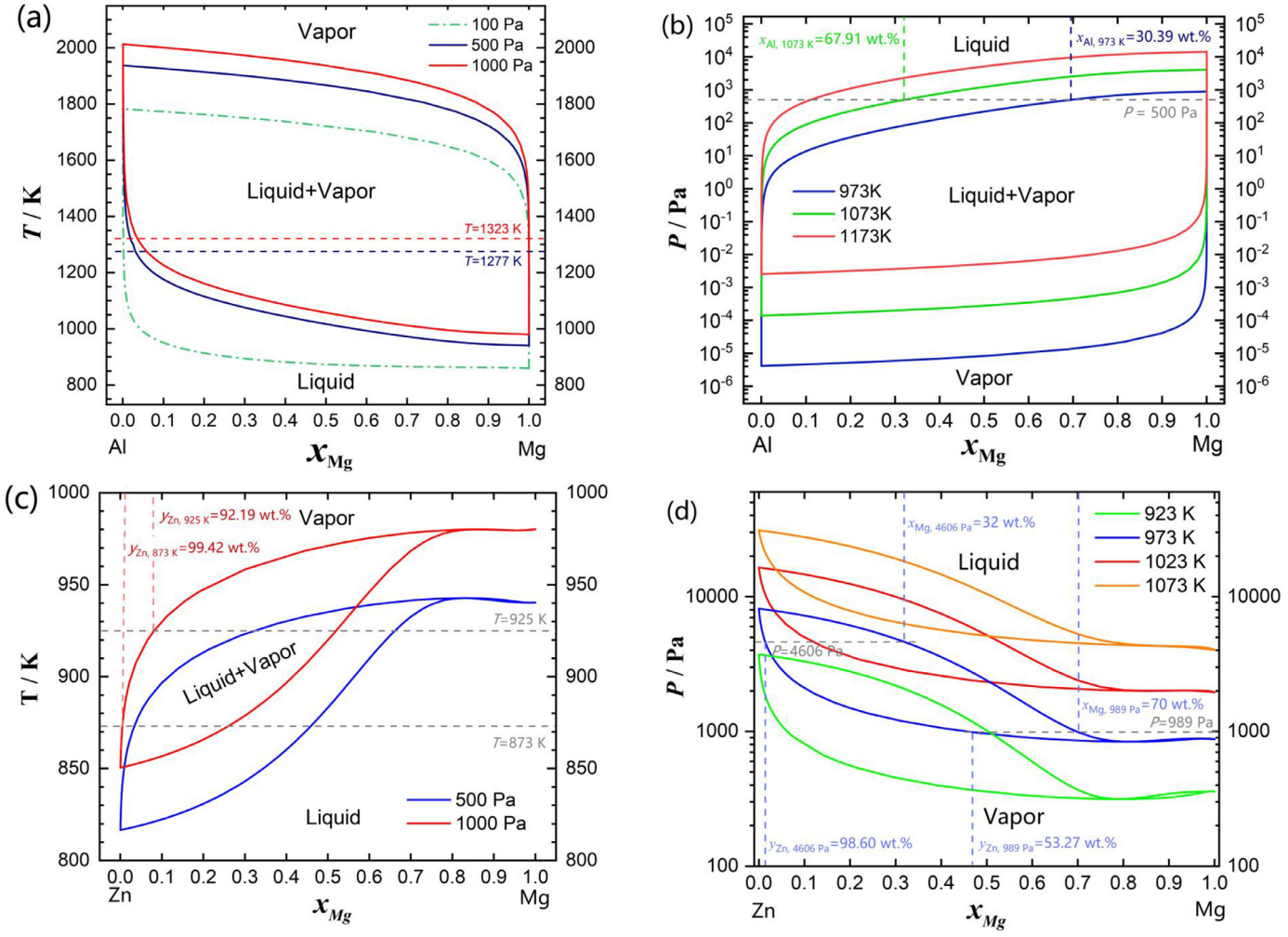


Fig. 6. VLE phase diagrams of the Mg-Al system:(a) *T-x(y)* and (b) *P-x(y)*; Mg-Zn system: (c)*T-x(y)* and (d) *P-x(y)*.

In the VLE phase diagrams in Fig. 6, the line near the vapor phase region indicates the Mg content in the vapor phase, while the line near the liquid phase region indicates the Mg content in the liquid phase. In a Mg-*i* binary system, the content of an impurity metal in the vapor and liquid phases can be calculated by equations (9) and (10). As shown in Fig. 6(a), in the *T-x(y)* phase diagram of Mg-Al, the coexistence region of the vapor and liquid phase decreased upon decreasing the system pressure and moved to a lower temperature. This indicates that the system pressure decreases result in the distillation temperature decreases. In addition, the optimal distillation temperature during the vacuum distillation separation of Mg-Al according to the purity requirements of distillation products can be obtained from the *T-x(y)* phase diagram. If the Mg content in the vapor phase (the purity of recycled Mg) is demanded to be greater than 99.99 wt.% when the system pressure is 1000 Pa, the distillation temperature must be lower than 1323 K and higher than 980 K. This is because a higher temperature will increase the volatility of Al and thus reduce the Mg content in the vapor phase. When the system pressure was reduced to 500 Pa, the distillation temperature range was reduced to 1277–940 K. Hence, reducing the system pressure can reduce the distillation temperature, thus reducing the energy consumption of the recovery process while ensuring good Mg and Al separation.

It should also be noted that the Mg alloy scrap recycling process focuses on the Mg-rich areas in the right of Fig. 6(b), while the Al content in the liquid phase will be higher after Mg recycling. If further recovery of Al is required, reducing the system pressure below 333 Pa can significantly reduce the distillation temperature (Al-rich region in the green dotted line on the left in Fig. 6(a)). This is because the pressure corresponding to the Al triple point is much lower than Mg, and Al remained in the liquid phase. More importantly, the use of vacuum distillation to recover and purify Al has been verified by several experimental studies [72–74].

Fig. 6(b) shows that at the same system pressure, upon increasing the temperature, the Mg content in the liquid phase gradually decreased, that is the purity of Al in the liquid phase increased. This implies that a high distillation temperature promoted the separation of Al and Mg during vacuum distillation. For example, when the temperature increased from

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Table 2
AM50A recycling experiment result by vacuum distillation.

		Impurity concentration (wt.%)									
		Al	Zn	Mn	Fe	Si	Cu	Ni	Pb	Sn	Ti
AM50A (Raw material)		4.9900	0.0800	0.3100	0.0018	0.0320	0.0019	0.0009	(-)	(-)	(-)
Recycled Mg of differed tray	#1	2.9110	0.0366	0.1822	0.0015	0.0105	0.0013	0.0003	0.0010	0.0001	0.0001
	#2	0.0047	0.0522	0.0005	0.0002	0.0019	0.0001	0.0001	0.0003	0.0001	0.0001
	#3	0.0007	0.0646	0.0002	0.0004	0.0017	0.0001	0.0001	0.0002	0.0001	0.0001
	#4	0.0005	0.0846	0.0001	0.0004	0.0017	0.0001	0.0001	0.0001	0.0001	0.0001
	#5	0.0005	0.0982	0.0001	0.0003	0.0015	0.0001	0.0001	0.0001	0.0001	0.0001
	#6	0.0007	0.0754	0.0001	0.0001	0.0015	0.0001	0.0001	0.0001	0.0001	0.0001
	#7	0.0005	0.1529	0.0001	0.0002	0.0017	0.0001	0.0001	0.0002	0.0001	0.0001
4 N Mg standard [76]		0.0020	0.0030	0.0020	0.0020	0.0020	0.0003	0.0003	0.0010	0.0020	0.0005
Residue		73.74	0.0002	8.416	0.01	0.1802	0.0224	0.0044	0.0187	0.0013	0.0001

(-) Indicates no detection because not required in relevant standard.

973 K to1073 K at 500 Pa, the Mg content in the vapor phase both was great than 99.99 wt.%, while the Al content in the liquid phase increased from 30.39 wt.% to 67.91 wt.%. Therefore, a lower pressure and higher temperature promoted the separation of Al, which can be applied to impurity metals with a separation coefficient similar to Al or even smaller.

For the Mg-Zn binary system, as shown in Fig. 6(c), the Zn content in the vapor phase increased from 92.19 wt.% to 99.42 wt.% as temperature decreased from 925 K to 873 K at 1000 Pa. This implies that a lower temperature facilitated the removal of zinc from magnesium by volatilization. In addition, when the system pressure was set to 500 Pa and 1000 Pa, the distillation temperature increased to 942 K and 980 K, respectively, and the vapor and liquid phase lines coincided in the *T-x(y)* phase diagram. This means that Mg and Zn could not be separated because they both evaporated too quickly under these conditions. Hence, it is showed again that a low temperature favored the separation of Zn from Mg, which is consistent with the trend in which the different saturation vapor pressures of Mg and Zn decreased upon increasing the temperature.

Fig. 6(d) shows the relationship between product composition, distillation temperature, and pressure during vacuum distillation. As the system pressure increased, the Zn content in the vapor phase increased. For example, when the system pressure increased from 989 Pa to 4606 Pa at 973 K, the purity of Mg in the liquid phase decreased from 70 wt.% to 32 wt.%, and the Zn content in the vapor phase increased from 53.27 wt.% to 98.60 wt.%, indicating that a high pressure facilitated the separation of Zn from Mg. This is well applicated and verified that zinc could be removed to prepare high-purity magnesium [75]. Therefore, properly increasing the system pressure promotes the separation of Mg and Zn during recycling.

3.3. Application of Mg alloy scrap recycling by vacuum distillation

A Mg alloy manufacturer in Shanxi Province adopted the vacuum distillation process to recycle AM50A scrap using a distillation column with staggered trays at 1023 K and 1000 Pa. The tray order was sorted from bottom (#1) to top (#7). The recycling case results confirmed the thermodynamic analysis, as shown in Table 2.

The recycling results of trays #3 to #7 in Table 2 demonstrate that vacuum distillation separated impurity metals except for Zn from the Mg melt. The content of impurity metals except Zn in the recycled Mg meets the standard of 4 N high-purity Mg [76], where the contents of Cu, Fe, Ni, Sn, and Ti were decreased to 1–4 ppm. However, Zn was present in all trays, and its content increased as the tray temperature decreased, which was determined by the condensation temperature of Zn. Therefore, zinc removal was the most critical factor affecting the purity of recycled magnesium using the vacuum distillation recycling process. The contents of Al, Mn, Pb, and Si in trays #1 and #2 increased significantly due to the continuous enrichment of impurity metals in the residue during the later stage of distillation. This rapidly increased their contents in the vapor phase, especially Al, which showed the highest content in the liquid phase. This conclusion seems to contradict the theoretical analysis that the Al content in the vapor phase was less than 1 ppm, but during the actual recycling process, Al vapor constantly condensed and accumulated in recycled Mg, especially during the later stage, resulting in an abnormal increase in Al content in tray #1. This also explains the results of Mn and Pb. Therefore, reasonably controlling the distillation process to avoid the contamination of Al, Mn, and other elements is necessary.

The Si content in recycled Mg was surprisingly high, even when its content was less than 20 ppm, which meets the standard of 4 N high-purity Mg. The Si content was higher than that of Cu, Fe, and Ni, which contradicts the separation coefficient of Si ($\lg\beta_{Si}= -16.851$) being smaller than that of Cu ($\lg\beta_{Cu}= -10.644$), Fe ($\lg\beta_{Fe}= -8.541$), and Ni ($\lg\beta_{Ni}= -15.020$). This discrepancy may be attributed to the reaction between excess Mg and Si, which formed Mg_2Si during distillation and was carried into the vapor phase. The formation of Mg_2Si can occur at approximately 823 K, and because the atomic mass of Si (28.085 g/mol) is not very

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different from that of Mg (24.305 g/mol), Mg_2Si can be easily carried into Mg vapor.

Based on the above thermodynamic analysis and experimental cases from Mg alloy manufacturers, several processing strategies can be proposed to improve vacuum distillation for the sustainable recycling of complex Mg alloy scrap. To ensure the purity of recycled Mg, a high pressure should be used to separate Zn from the Mg melt at low temperatures during the early stage of distillation. Subsequently, a lower system pressure and temperature should be used to efficiently separate other impurity metals from Mg. However, zinc removal is still challenging during the recovery of magnesium alloy scrap via vacuum distillation, and further research is needed. In addition, while economically feasible, the alloying elements Al and Zn can also be recovered and purified by vacuum distillation to maximize the recycling of metal resources.

4. Conclusions

A thermodynamic evaluation of the sustainable recycling of possible inclusions and 23 elements in complex Mg alloy scrap during vacuum distillation has been investigated systematically and comprehensively, and it validated by recovery experiment. The following conclusions were drawn:

(1) Non-metallic inclusions were stabilized and remained in the residue during vacuum distillation recycling. Chloride inclusions were volatile, but their content was limited and did not affect the recovery of recycled Mg.
(2) The separation coefficient $\lg\beta_i = -5$ is a thermodynamic criterion for evaluating the separation and removal capacity of impurity metals from Mg by vacuum distillation. Cu, Fe, Co, Ni, and other metals with $\lg\beta_i \leq -5$ could be removed, while Li, Sr, Ca, Mn, and Pb with $-1 < \lg\beta_i < -5$ may contaminate the recycled Mg. Zn with $\lg\beta_i \geq -1$ was almost impossible to remove.
(3) An optimization strategy for vacuum distillation recycling Mg alloy scrap: a low temperature and high system pressure during the early stage will promote the separation of Zn from Mg. During the later stage, this process can be terminated early, or a continuous supply of feedstock melt can be used to avoid contamination by Al, Mn, Ca, and Li in the liquid phase.
(4) The alloying elements Al and Zn in Mg alloy scrap can be further recovered and purified by vacuum distillation in theory when economically feasible, and it need to further research to maximize recycling and sustainable metal resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Lipeng Wang: Conceptualization, Data curation, Methodology, Writing – original draft, Investigation. **Dong Liang:** Formal analysis, Methodology, Visualization. **Yang Tian:** Investigation, Project administration, Supervision, Writing – review & editing. **Jianxue Chai:** Data curation, Supervision. **Rui Li:** Data curation, Investigation, Visualization. **Shuji Wu:** Data curation, Visualization, Investigation. **Bin Yang:** Conceptualization, Supervision, Writing – review & editing. **Baoqiang Xu:** Conceptualization, Methodology, Supervision. **Yong Deng:** Methodology, Validation, Visualization.

Acknowledgments

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

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Please cite this article as: L. Wang, D. Liang, Y. Tian et al., Thermodynamic and experimental evaluation of the sustainable recycling of magnesium alloy scrap by vacuum distillation based on vapor-liquid equilibrium, Journal of Magnesium and Alloys, https://doi.org/10.1016/j.jma.2023.12.011